Jack A. Logan

Theoretical and Computational Physicist

Mathematical Modeling | Numerical Algorithms | Scientific Computing

Salt Lake City, UT | Open to relocation | jalogan.github.io | LinkedIn | GitHub

Research Profile

Theoretical and computational physicist with a PhD and 9+ years of experience developing original mathematical models, numerical algorithms, and scientific software for complex physical systems. Led theoretical and computational research on statistical mechanics, interacting particles, symmetry-resolved structure, and molecular simulation, with validation against numerical and experimental results. Built custom C++ simulation tools and Python analysis methods; additional experience in physics-informed neural-network models. First/co-first author on five peer-reviewed papers. Interested in research roles involving mathematical model development, scientific algorithms, and computational discovery.

Technical Skills

Scientific modeling: Statistical mechanics | molecular dynamics | Monte Carlo | symmetry and tensors | constrained numerical integration | computational geometry | asymptotic modeling

Programming and ML: Python | C++ | NumPy/SciPy | PyTorch | scikit-learn | LAMMPS | Git | Linux | HPC/parallel computing | CuPy (CUDA GPU acceleration)

Education

Stony Brook University, Stony Brook, NY — PhD Physics 2022; MA Physics 2016; BS Physics 2009

Long Island University, Brookville, NY — MS Applied Mathematics 2014, *Summa Cum Laude*

PhD Dissertation: *Geometric and Topological Aspects of Self-Assembly: from spheres and rods to designer particles*

Research Experience

Yale University, New Haven, CT 2022-2025

Postdoctoral Fellow, Mechanical Engineering & Materials Science

- Led development of six C++ coarse-grained protein model variants, investigating backbone geometry, angle/dihedral constraints, and sequence-dependent side-chain representations including modified Martini-inspired models.
- Selected and curated 2,531 experimentally characterized proteins, developed quantitative structural observables, and performed 15,186 model–protein simulation runs (2,531 for each of six models) to identify the physical constraints needed to reproduce experimental structure.
- Implemented CuPy GPU acceleration for structure-factor calculations and a custom Monte Carlo analysis estimating core packing fractions for irregularly shaped amino acid models.
- Led the scientific modeling effort and worked with student researchers; results published as first author *Physical Review E* Editors' Suggestion (2025).

Brookhaven National Laboratory, Upton, NY 2016-2022

Research Assistant, Center for Functional Nanomaterials

Doctoral research at BNL during Stony Brook University graduate study (overlapping dates).

- Developed original statistical-mechanical theories and computational algorithms for constrained sphere packings, particle interactions, and symmetry-adapted structural characterization, including topology-changing Monte Carlo methods and analytical interaction potentials.
- Led theoretical and computational analysis in multi-institution research collaborations, developing mathematical frameworks, structural classification algorithms, and quantitative methods to characterize and validate ordered and disordered nanoparticle structures against simulations and experiments.

Selected Research Contributions

Symmetry-Specific Bond Order Parameters (SymBOPs)

Mathematical descriptors | Group-theoretic representations | Data analysis

- Developed symmetry-adapted tensor descriptors and bond-level classification and domain-detection algorithms to identify crystalline domains, coherent amorphous order, and structural defects; validated methods using simulated and experimental nanoparticle data across three peer-reviewed studies.

Statistical Mechanics of Sphere Packing
Original theory | Monte Carlo | Rigidity and zero modes
- Developed a theory separating geometric and topological entropy and an isostaticity-preserving Monte Carlo algorithm for sampling packing topologies through zero-mode motion; implemented pivoted QR to solve constrained linear systems.

Coarse-Grained Protein Modeling
Scientific C++ | HPC | Experimental benchmarking
- Led development of six coarse-grained protein models and analysis of 15,186 simulations across 2,531 proteins, using object-oriented C++ simulations and GPU-accelerated structural analysis to identify stereochemical constraints governing key structural properties of experimentally determined folded proteins.

Analytical Interaction Potential for Nanorods
First-principles derivations | Asymptotics | Computational efficiency
- Derived and numerically validated efficient analytical approximations for finite-nanorod interactions at arbitrary positions and orientations, replacing costly multidimensional integrations while capturing near- and far-field behavior.

Hybrid Isotropic/Patchy-Particle Self-Assembly
Research leadership | Morphology | Structure identification
- Led theoretical and structural analysis of a collaborative self-assembly study, developing symmetry-based characterization and scientific visualizations to identify crystalline and coherent amorphous morphologies in collaborator-generated simulations.

Additional Scientific Computing Projects

Curved-Surface Molecular Dynamics
Custom C++ engine | RATTLE constraints | Geometry
- Built a custom C++ molecular-dynamics engine using RATTLE constraints for 3D spheres tangent to planar and curved substrates; implemented surface-aware interactions, jamming procedures, and surface-meshing algorithms.

Physics-Informed Neural-Network Force Field
PyTorch | Dynamical modeling | Validation
- Developed a PyTorch Lennard–Jones force model preserving rotational equivariance and action–reaction symmetry; integrated it into molecular dynamics and evaluated force errors, trajectory divergence, and structural observables.

Mechanistic Analysis of D_3 Symmetry in a Transformer
PyTorch | Group theory | Activation interventions
- Trained a Transformer on dihedral-group operations and developed activation-reinjection experiments to investigate learned composition, analyzing internal representations using sparse autoencoders, hidden-state similarities, and PCA.

Selected Publications

- **Logan*** et al., "Effect of stereochemical constraints on the structural properties of folded proteins," *Phys. Rev. E* **112**, 054405 (2025). **Editors' Suggestion; equal first author**.
- **Logan** et al., "Symmetry-specific characterization of bond orientation order in DNA-assembled nanoparticle lattices," *J. Chem. Phys.* **159**, 154905 (2023).
- **Logan** et al., "Symmetry-specific orientational order parameters for complex structures," *J. Chem. Phys.* **156**, 054108 (2022).
- **Logan** and Tkachenko, "Geometric and topological entropies of sphere packing," *Phys. Rev. E* **105**, 014117 (2022).
- Mushnoori, **Logan**, Tkachenko, and Dutt, "Controlling morphology in hybrid isotropic/patchy particle assemblies," *J. Chem. Phys.* **156**, 024501 (2022), **Editors' Pick**.
- **Logan** and Tkachenko, "Compact interaction potential for van der Waals nanorods," *Phys. Rev. E* **98**, 032609 (2018).

Additional peer-reviewed publications include protein-core jamming (*PRX Life*, 2025) and near-field spectroscopy (*Nature Communications*, 2017).

Selected Research Leadership & Awards

- Led modeling and theoretical analysis in multi-institution research collaborations involving experimental scientists and simulation groups.
- Team Leader, Protein Folding Simulation Project, Yale University; mentored student summer researchers.
- GSNP Postdoctoral Speaker Award Finalist, APS March Meeting; Jonathan Kaufman Student Excellence Prize.
- Yale postdoctoral and Physical and Engineering Biology (PEB) research travel awards.